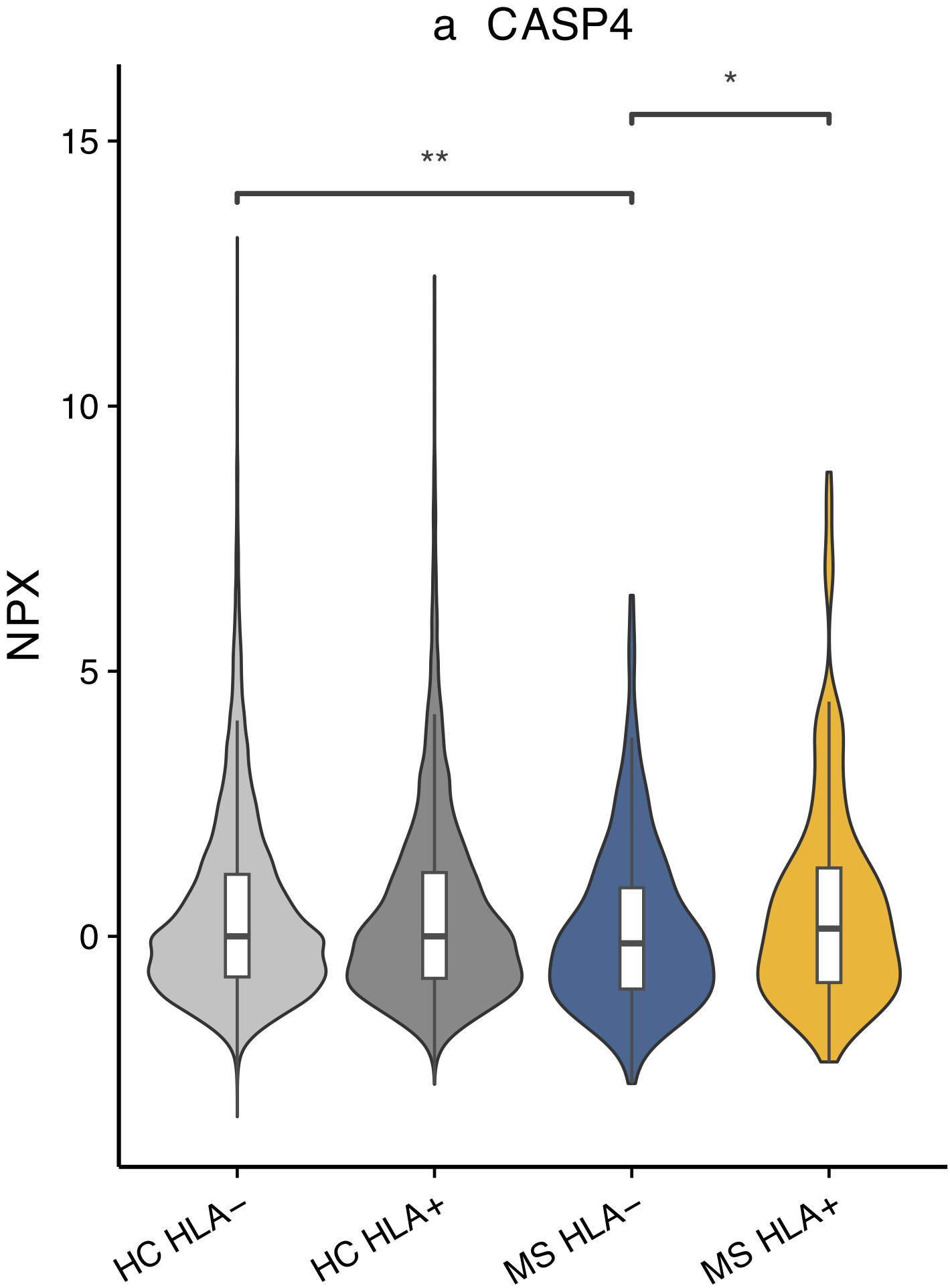
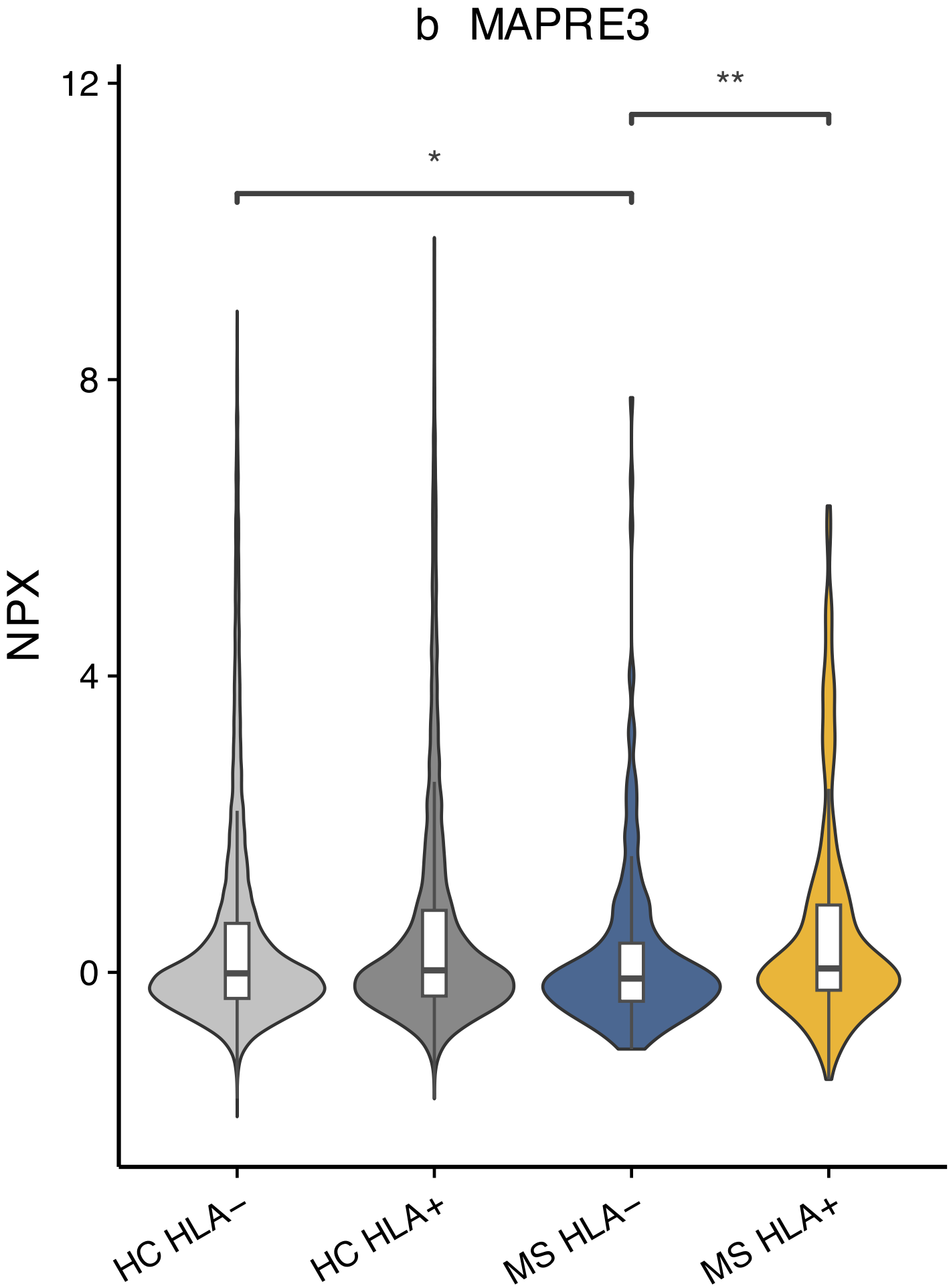
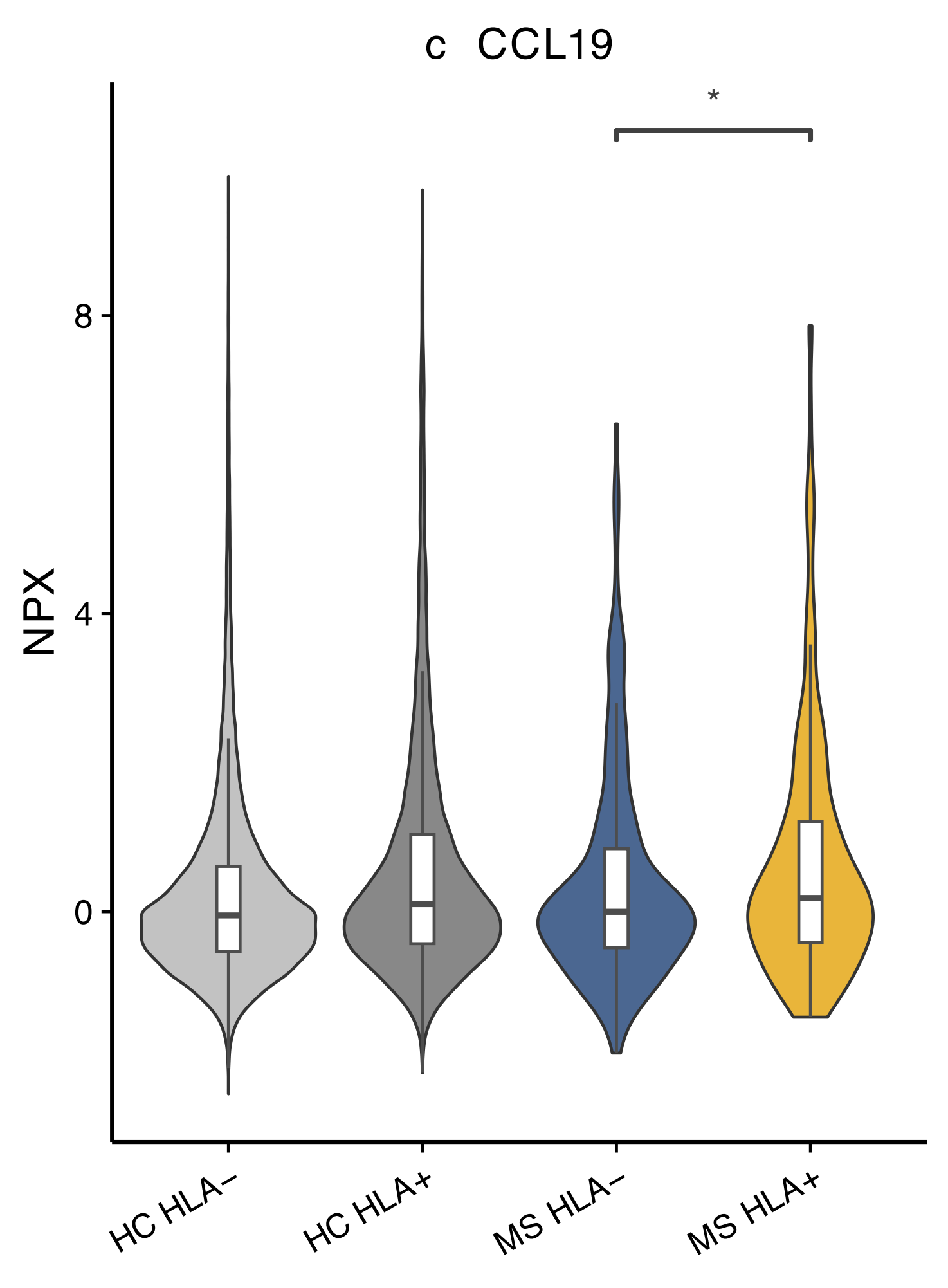
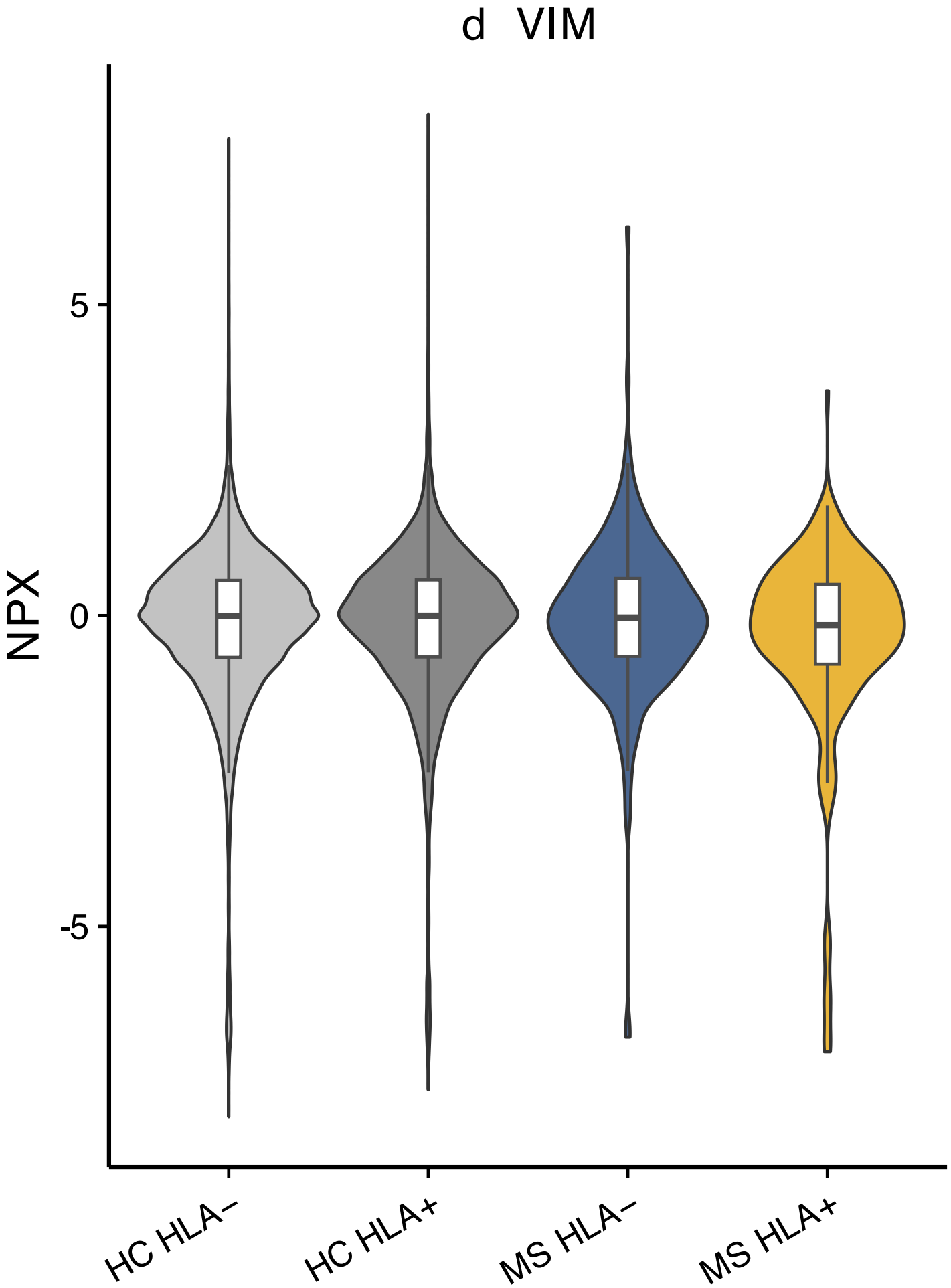
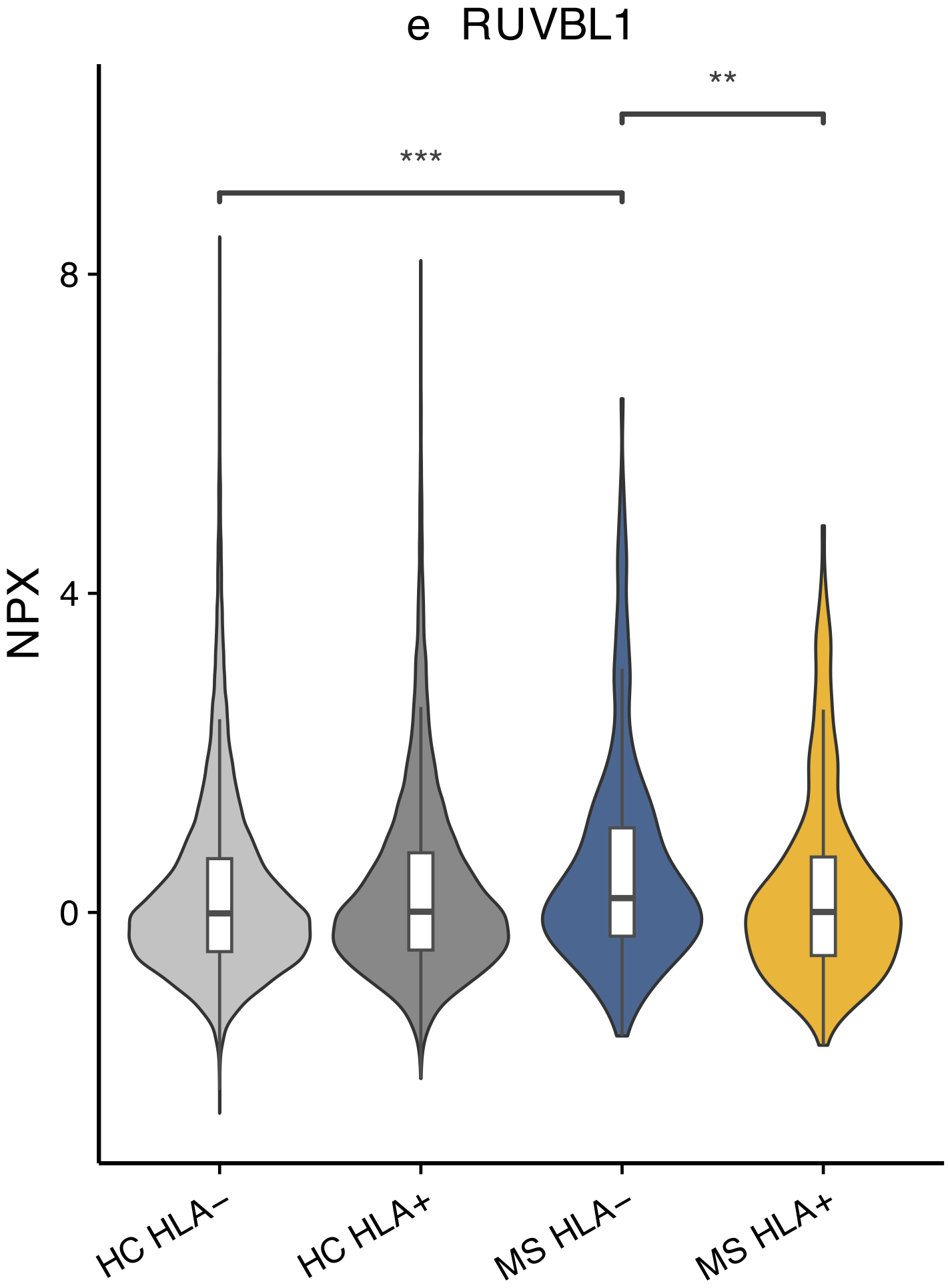
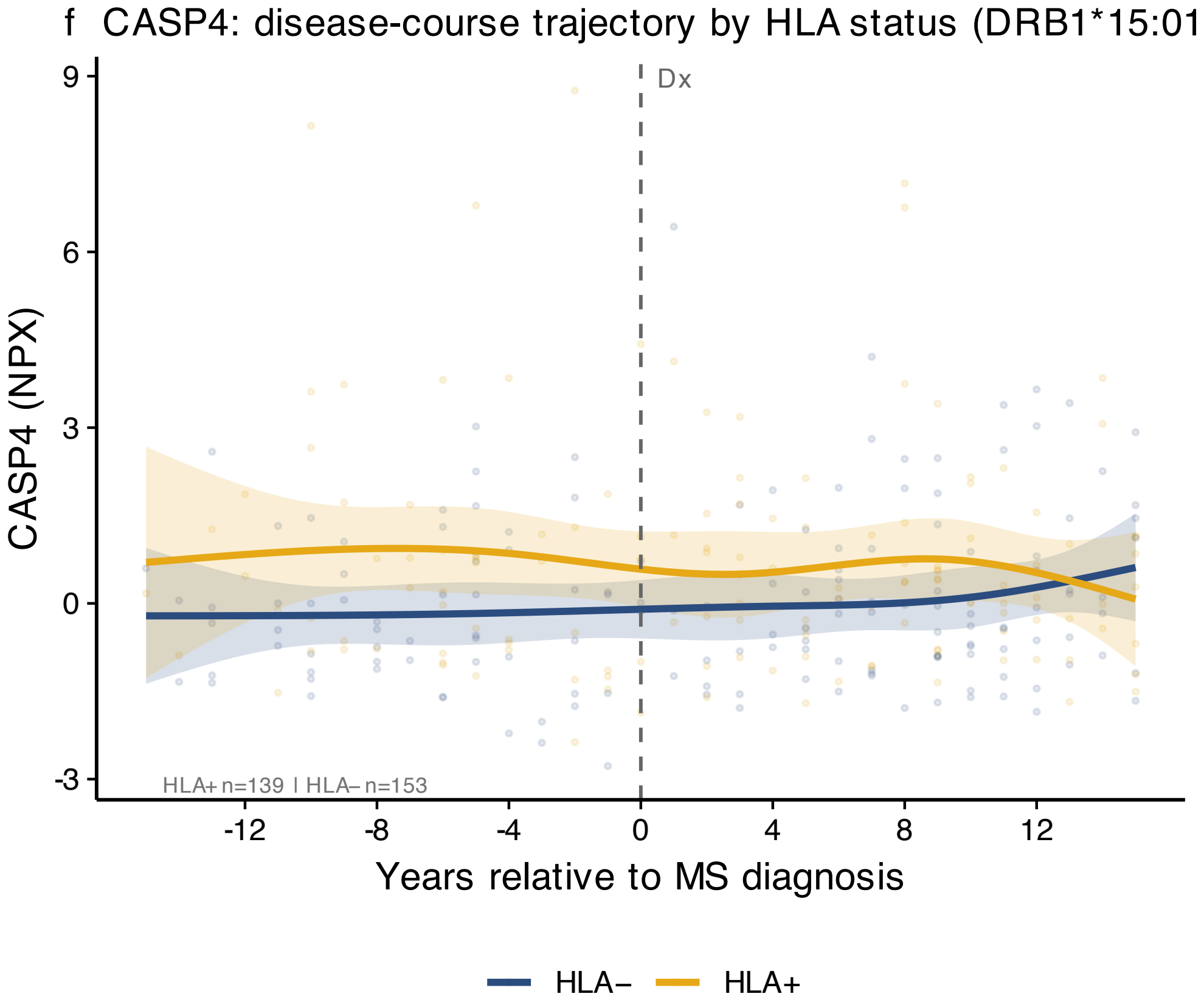

a CASP4
NPX
15
10
5
0
**
*
HC HLA−
HC HLA+
MS HLA−
MS HLA+

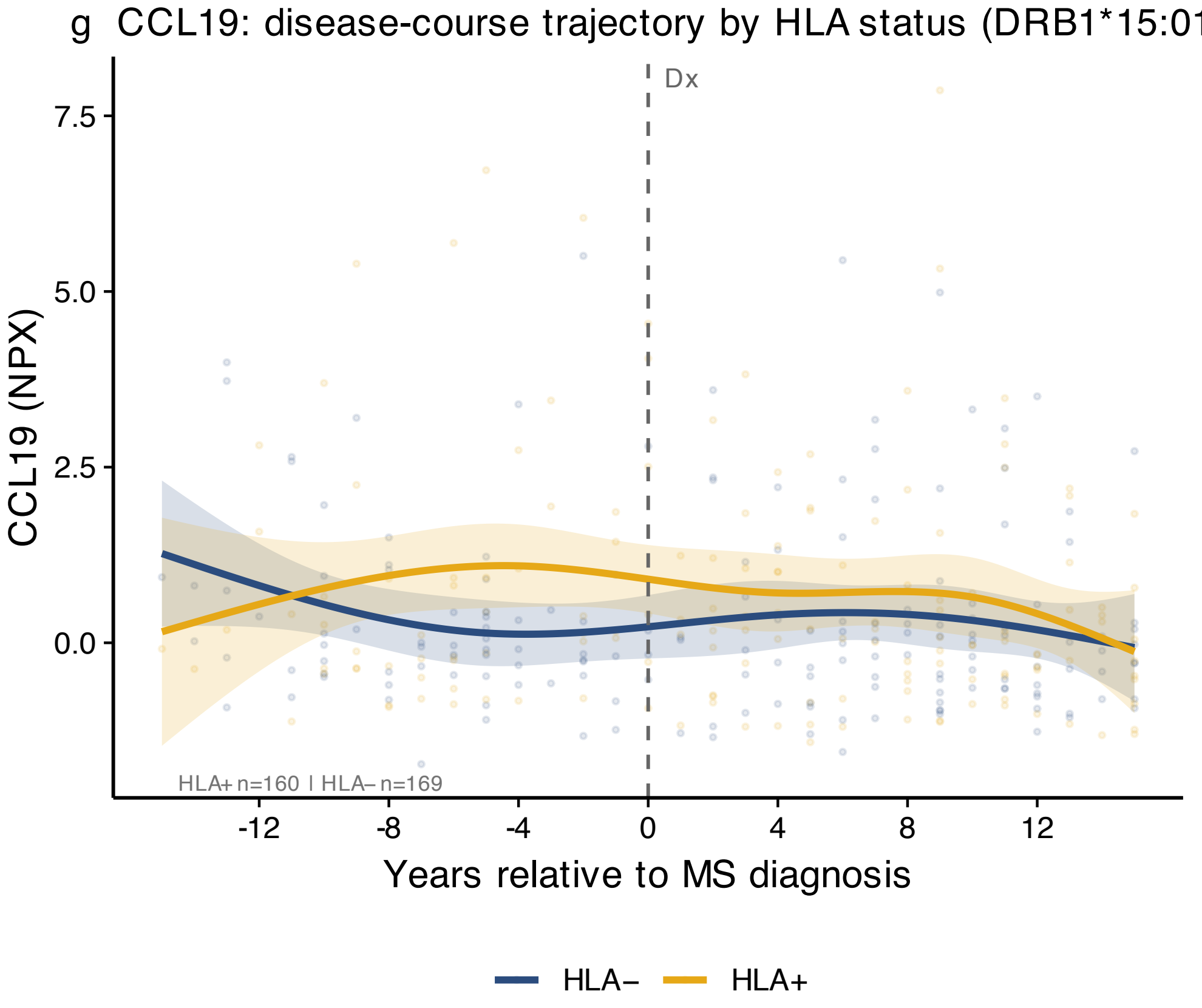

b MAPRE3
NPX
12
8
4
0
*
**
HC HLA−
HC HLA+
MS HLA−
MS HLA+

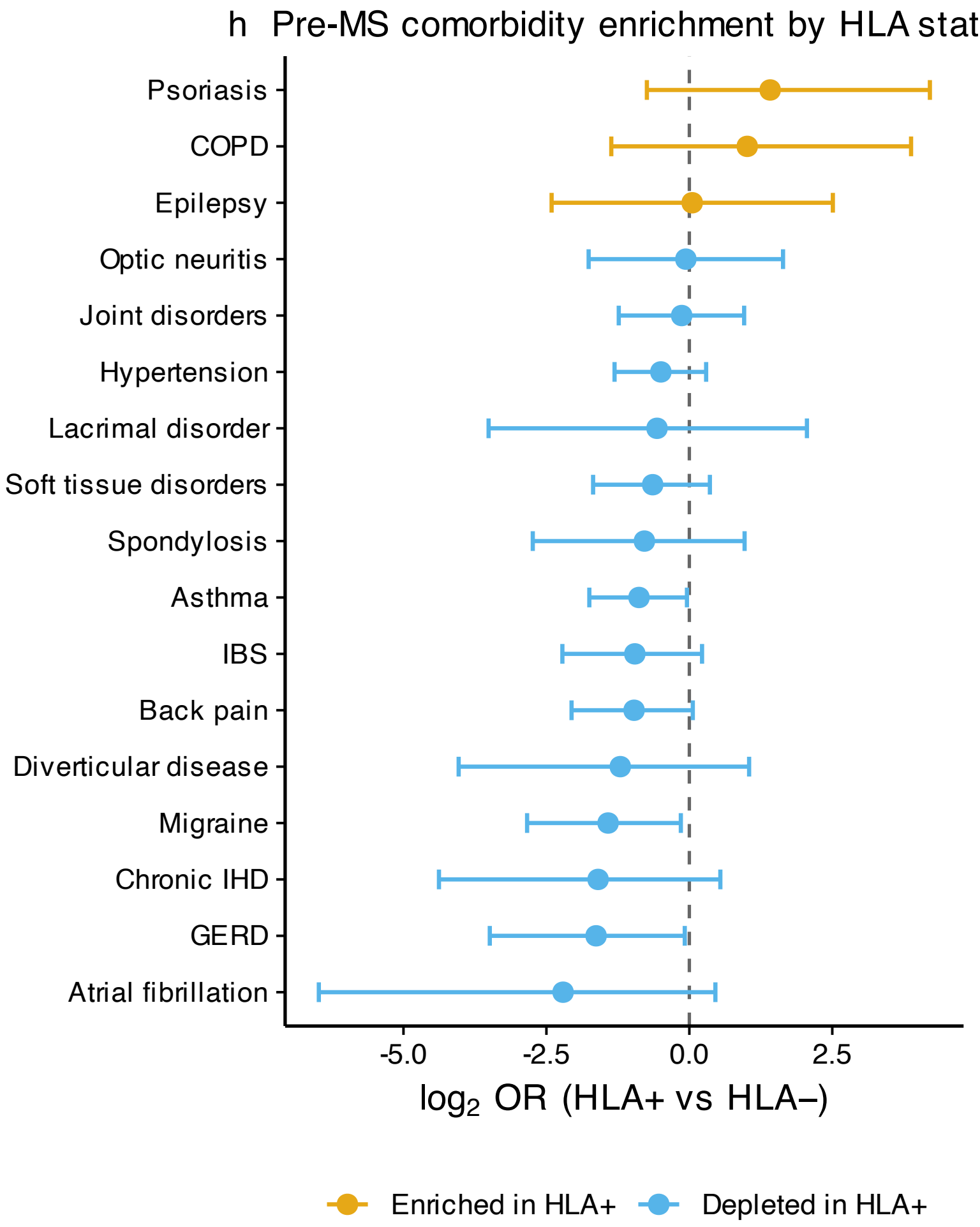

c CCL19
*
8
4
0
NPX
HC HLA−
HC HLA+
MS HLA−
MS HLA+

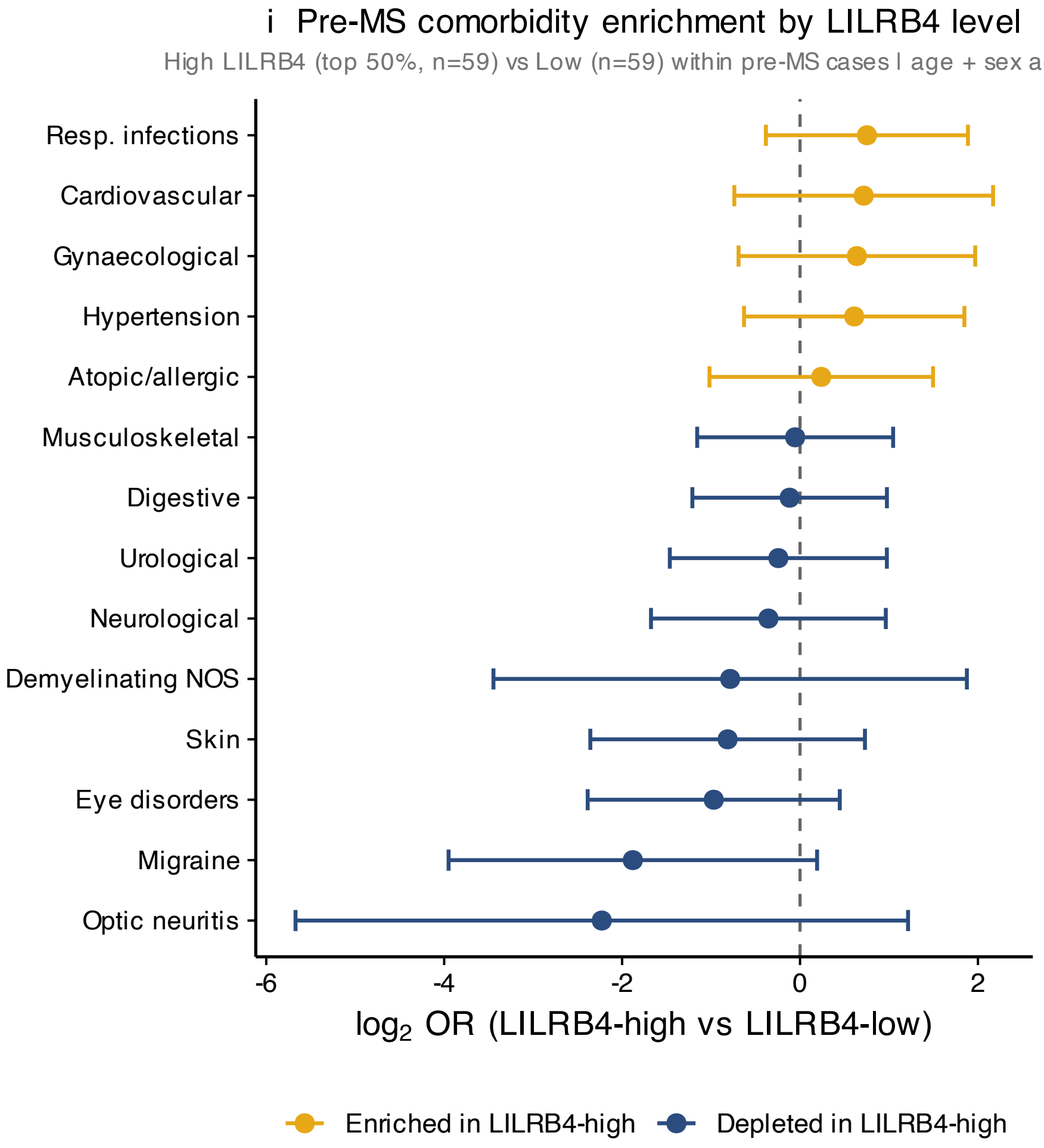

d VIM
NPX
5
0
-5
HC HLA−
HC HLA+
MS HLA−
MS HLA+

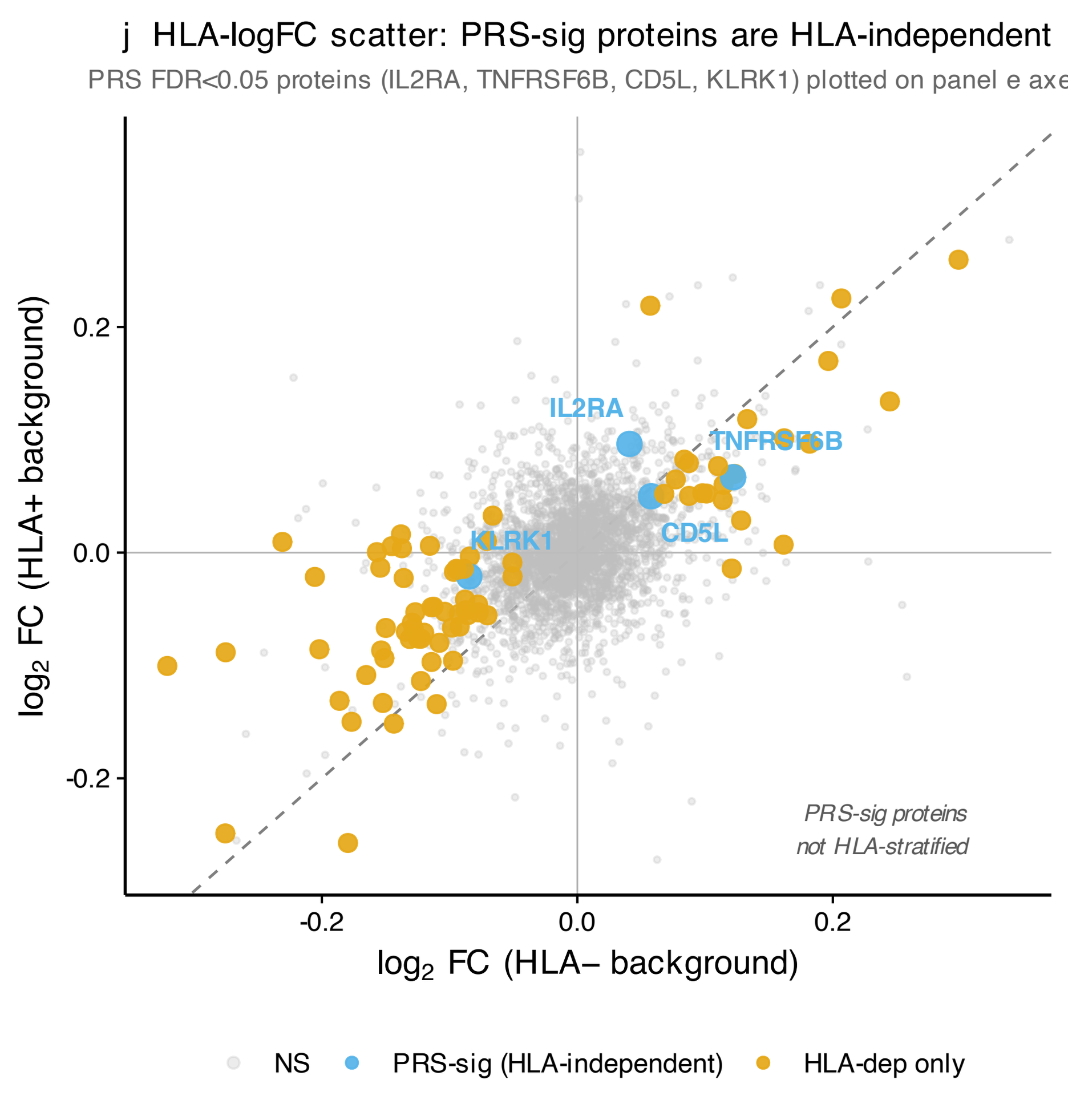

e RUVBL1
**
***
8
4
0
NPX
HC HLA−
HC HLA+
MS HLA−
MS HLA+


f CASP4: disease-course trajectory by HLA status (DRB1*15:01
Dx
CASP4 (NPX)
9
6
3
0
-3
HLA+ n=139 | HLA− n=153
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis
HLA−
HLA+


g CCL19: disease-course trajectory by HLA status (DRB1*15:01
Dx
7.5
5.0
2.5
0.0
CCL19 (NPX)
HLA+ n=160 | HLA− n=169
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis
HLA−
HLA+


h Pre-MS comorbidity enrichment by HLA stat
Psoriasis
COPD
Epilepsy
Optic neuritis
Joint disorders
Hypertension
Lacrimal disorder
Soft tissue disorders
Spondylosis
Asthma
IBS
Back pain
Diverticular disease
Migraine
Chronic IHD
GERD
Atrial fibrillation
-5.0
-2.5
0.0
2.5
log2 OR (HLA+ vs HLA–)
Enriched in HLA+
Depleted in HLA+


i Pre-MS comorbidity enrichment by LILRB4 level
High LILRB4 (top 50%, n=59) vs Low (n=59) within pre-MS cases | age + sex ad
Resp. infections
Cardiovascular
Gynaecological
Hypertension
Atopic/allergic
Musculoskeletal
Digestive
Urological
Neurological
Demyelinating NOS
Skin
Eye disorders
Migraine
Optic neuritis
-6
-4
-2
0
2
log2 OR (LILRB4-high vs LILRB4-low)
Enriched in LILRB4-high
Depleted in LILRB4-high


j HLA-logFC scatter: PRS-sig proteins are HLA-independent
PRS FDR<0.05 proteins (IL2RA, TNFRSF6B, CD5L, KLRK1) plotted on panel e axe
IL2RA
TNFRSF6B
CD5L
KLRK1
PRS-sig proteins
not HLA-stratified
0.2
0.0
-0.2
-0.2
0.0
0.2
log2 FC (HLA+ background)
log2 FC (HLA– background)
NS
PRS-sig (HLA-independent)
HLA-dep only